

Junzhe Shen

CONTACT INFORMATION

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RESEARCH INTERESTS

Research Areas: LLM Pretraining; Concept-Level Language Modeling; Diffusion Language Models.

I aim to develop stronger pretraining paradigms for foundation language models, especially objectives and representations beyond vanilla next-token prediction:

- **New pretraining architectures for LLMs:** designing model architectures and harder pretraining tasks that provide richer supervision than vanilla next-token prediction.
- **Efficient and controllable language modeling:** diffusion and non-autoregressive LMs with improved training objectives, noise schedules, parameterizations, and architectures.

EDUCATION

Shanghai Jiao Tong University 2025 – 2030 (expected)
School of Computer Science
Ph.D. Candidate, Advisor: Prof. Zhouhan Lin

Shanghai Jiao Tong University 2021 – 2025
School of Computer Science
Information Security (IEEE pilot class)

INTERNSHIP EXPERIENCE

Beijing Institute for General Artificial Intelligence Sep. 2024 – May. 2025
Student Researcher

ONGOING RESEARCH

Concept-Level Pretraining Objectives for Foundation Language Models Ongoing
Unpublished research related to Next Concept Prediction in Discrete Latent Space Leads to Stronger Language Models

- Investigating objectives that predict latent concepts instead of only next tokens, exploring discrete concept representations and continual pretraining recipes for stronger LLM pretraining.

Syntactic-Hierarchy-Aware Masked Diffusion Language Models Ongoing
Unpublished research on improving masked diffusion language modeling

- Incorporating syntactic hierarchy into masking and denoising strategies to provide structure-aware supervision and improve masked diffusion LM performance.

PUBLICATIONS

- Tong Wu*, **Junzhe Shen***, Zixia Jia, Yuxuan Wang, Zilong Zheng. TokenSwift: Lossless Acceleration of Ultra Long Sequence Generation. ICML, 2025
- Zixia Jia, Jiaqi Li, Yipeng Kang, Yuxuan Wang, Tong Wu, Quansen Wang, Xiaobo Wang, Shuyi Zhang, **Junzhe Shen**, Qing Li, Siyuan Qi, Yitao Liang, Di He, Zilong Zheng, Song-Chun Zhu. The AI Hippocampus: How Far are We From Human Memory? TMLR, 2026
- **Junzhe Shen**, Jieru Zhao, Ziwei He, Zhouhan Lin. CoDAR: Continuous Diffusion Language Models are More Powerful Than You Think. <https://arxiv.org/abs/2603.02547>

HONORS AND AWARDS

- Academic Scholarship C (twice)
- Second Prize, National College Student Information Security Competition (CISCN)